COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	10/29/2026	**DATE SUBMITTED:**	10/29/2026
NAME:	Charles Vence Espinas	**INSTRUCTOR:**	Engr. Jimlord Quejado

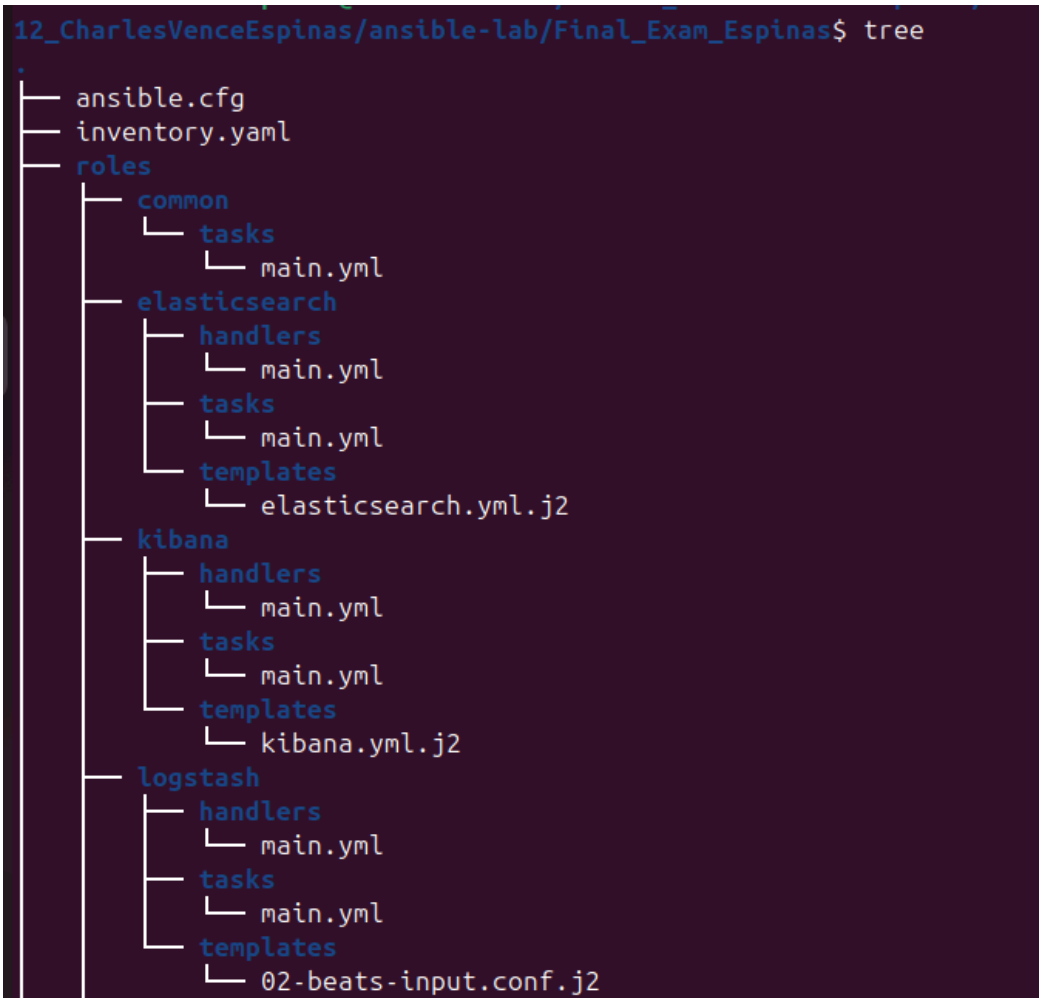
FINAL HANDS-ON SKILLS EXAM

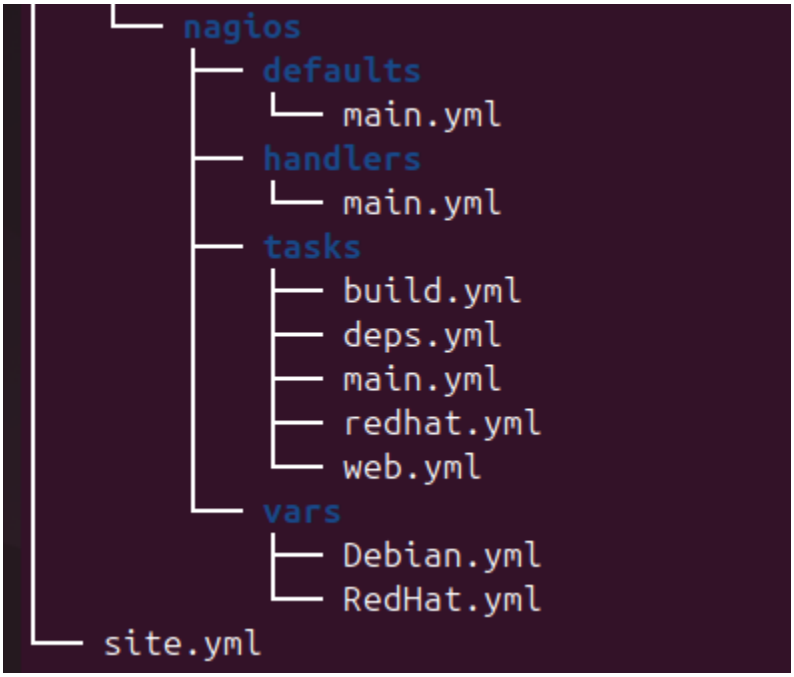
1. PROCEDURE OUTPUT

Tools Needed
1. VM with Ubuntu, CentOS and Ansible installed
2. Web browser
Github Link: https://github.com/charlesespinas/CPE212_CharlesVenceEspinas

Procedure
1. Create a repository and label it as "Final_Exam_Surname"
2. Clone your new repository in your VM
3. Create an Ansible playbook that does the following with an input of a config.yaml file and structure inventory file.
3.1 Install and configure one enterprise service that can be installed in Debian and Centos servers
3.2 Install and configure one monitoring tool that can be installed in Debian and Centos servers (if it is a stack there should be option of different host)
4.4 Change Motd as "Ansible Managed by <username>"
4. Push and commit your files in GitHub
5. Make sure to show evidence of input (codes) process (codes successfully running) and output (evidence of installation)
5. For your final exam to be counted, please paste your repository link as an answer in this exam.
Note: Extra points if you will implement the said services via containerization.

File Structure





ansible configuration

```
  GNU nano 7.2                                          ansible.cfg
[defaults]
inventory = inventory.yaml
roles_path = roles
host_key_checking = False
interpreter_python = auto_silent
retry_files_enabled = False
motd_username: "Charles"

[privilege_escalation]
become = True
become_method = sudo
```

inventory

```
  GNU nano 7.2                                          inventory.yaml
all:
  vars:
    ansible_user: charles-vence-espinas

  children:
      Ubuntu_servers:
        hosts:
          server1:
            ansible_host: 192.168.121.9
            os_type: Ubuntu

          server2:
            ansible_host: 192.168.121.5
            os_type: Ubuntu

      CentOS_servers:
        hosts:
          centOS:
            ansible_host: 192.168.121.6
            os_type: CentOS

      linux_servers:
        children:
          Ubuntu_servers:
          CentOS_servers:
```

site.yml

```
  GNU nano 7.2                                                site.yml
---
- name: Install and Configure Nagios Core across Ubuntu and CentOS
  hosts: linux_servers
  become: true
  roles:
    - nagios

- name: Apply common configurations across all ELK nodes
  hosts: linux_servers
  become: true
  roles:
    - common

- name: Install and configure Elasticsearch
  hosts: linux_servers
  become: true
  roles:
    - elasticsearch

- name: Install and configure Logstash
  hosts: linux_servers
  become: true
  roles:
    - logstash
```

```
- name: Install and configure Kibana
  hosts: linux_servers
  become: true
  roles:
    - kibana

- name: Set MOTD message
  hosts: linux_servers
  become: true
  tasks:
    - name: Copy MOTD content
      copy:
        dest: /etc/motd
        content: "Ansible Managed by {{ motd_username }}\n"
        mode: '0644'
```

roles/common/tasks/main.yml

```yaml
---
- name: Create swap file on low-memory nodes (<= 2GB RAM)
  block:
    - name: Create 2GB swap file
      command: fallocate -l 2G /swapfile
      args:
        creates: /swapfile

    - name: Set correct permissions on swap file
      file:
        path: /swapfile
        owner: root
        group: root
        mode: '0600'

    - name: Make swap filesystem
      command: mkswap /swapfile
      register: mkswap_result
      changed_when: "'setting up swapspace' in mkswap_result.stdout"
      failed_when: false

    - name: Enable swap
      command: swapon /swapfile
      register: swapon_result
      changed_when: "swapon_result.rc == 0"
      failed_when: false
```

```yaml
    - name: Ensure swap is mounted on boot in fstab
      mount:
        name: none
        src: /swapfile
        fstype: swap
        opts: sw
        state: present
  when: ansible_memtotal_mb <= 2048

- name: Install prerequisites on Debian/Ubuntu
  apt:
    name:
      - apt-transport-https
      - gnupg2
      - openjdk-11-jre
    state: present
    update_cache: yes
  when: ansible_os_family == "Debian"

- name: Add Elastic GPG key on Debian/Ubuntu
  apt_key:
    url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    state: present
  when: ansible_os_family == "Debian"
```

```
- name: Import Elastic GPG key on RHEL/CentOS
  rpm_key:
    key: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    state: present
  when: ansible_os_family == "RedHat"

- name: Add Elastic repository on RHEL/CentOS
  yum_repository:
    name: elastic-8.x
    description: Elastic repository for 8.x packages
    baseurl: https://artifacts.elastic.co/packages/8.x/yum
    gpgcheck: yes
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    enabled: yes
  when: ansible_os_family == "RedHat"
```

roles/elasticsearch/tasks/main.yml

```
---
- name: Install Elasticsearch
  package:
    name: elasticsearch
    state: present

- name: Ensure Elasticsearch data and log directories exist
  file:
    path: "{{ item }}"
    state: directory
    owner: elasticsearch
    group: elasticsearch
    mode: '0755'
  loop:
    - /var/lib/elasticsearch
    - /var/log/elasticsearch

- name: Limit JVM heap size to 512MB
  copy:
    dest: /etc/elasticsearch/jvm.options.d/heap.options
    content: |
      -Xms512m
      -Xmx512m
    owner: elasticsearch
    group: elasticsearch
    mode: '0644'
```

```
- name: Configure elasticsearch.yml
  template:
    src: elasticsearch.yml.j2
    dest: /etc/elasticsearch/elasticsearch.yml
    owner: elasticsearch
    group: elasticsearch
    mode: '0660'
  notify: restart elasticsearch

- name: Enable and start Elasticsearch service
  systemd:
    name: elasticsearch
    enabled: yes
    state: started

- name: Allow Elasticsearch port 9200 on Debian/Ubuntu (UFW)
  ufw:
    rule: allow
    port: '9200'
    proto: tcp
    state: enabled
  when: ansible_os_family == "Debian"
```

```
- name: Allow Elasticsearch port 9200 on RHEL/CentOS (firewalld)
  firewalld:
    port: 9200/tcp
    permanent: yes
    state: enabled
    immediate: yes
  when: ansible_os_family == "RedHat"
```

roles/elasticsearch/handlers/main.yml

```
  GNU nano 7.2                          roles/elasticsearch/handlers/main.yml
---
- name: restart elasticsearch
  systemd:
    name: elasticsearch
    state: restarted
```

roles/elasticsearch/templates/elasticsearch.yml.j2

```
  GNU nano 7.2                                roles/elasticsearch/te
cluster.name: enterprise-elk-cluster
node.name: node-1

path.data: /var/lib/elasticsearch
path.logs: /var/log/elasticsearch

network.host: 0.0.0.0
http.port: 9200
discovery.type: single-node

xpack.security.enabled: false
xpack.security.enrollment.enabled: false
xpack.security.http.ssl.enabled: false
xpack.security.transport.ssl.enabled: false
```

roles/kibana/handlers/main.yml

```
  GNU nano 7.2                                      roles/kibana/hand
---
- name: restart kibana
  systemd:
    name: kibana
    state: restarted
```

roles/kibana/tasks/main.yml

```
  GNU nano 7.2                                    roles/kibana/tasks/main.y
---
- name: Install Kibana
  package:
    name: kibana
    state: present

- name: Configure kibana.yml
  template:
    src: kibana.yml.j2
    dest: /etc/kibana/kibana.yml
    owner: root
    group: kibana
    mode: '0660'
  notify: restart kibana

- name: Enable and start Kibana service
  systemd:
    name: kibana
    enabled: yes
    state: started

- name: Allow Kibana port 5601 on CentOS firewall
  firewalld:
    port: 5601/tcp
    permanent: yes
    state: enabled
    immediate: yes
  when: ansible_os_family == "RedHat"
```

roles/kibana/templates/kibana.yml.j2

```
  GNU nano 7.2                          roles/kibana/templates/kibana.yml.j2
server.port: 5601
server.host: "0.0.0.0"
elasticsearch.hosts: ["http://{{ hostvars[groups['elasticsearch_node'][0]]['ansible_host'] }}:9200"]
```

roles/logstash/handlers/main.yml

```
  GNU nano 7.2                                  roles/logstash/handlers/main.yml
---
- name: restart logstash
  systemd:
    name: logstash
    state: restarted
```

roles/logstash/tasks/main.yml

```
  GNU nano 7.2                                  roles/logstash/tasks/main.yml
---
- name: Install Logstash
  package:
    name: logstash
    state: present

- name: Configure Logstash beats input pipeline
  template:
    src: 02-beats-input.conf.j2
    dest: /etc/logstash/conf.d/02-beats-input.conf
    owner: logstash
    group: logstash
    mode: '0644'
  notify: restart logstash

- name: Enable and start Logstash service
  systemd:
    name: logstash
    enabled: yes
    state: started
```

roles/logstash/templates/02-beats-input.conf.j2

```
  GNU nano 7.2                          roles/logstash/templates/02-beats-input.conf.j2
input {
  beats {
    port => 5044
  }
}

output {
  elasticsearch {
    hosts => ["http://{{ hostvars[groups['elasticsearch_node'][0]]['ansible_host'] }}:9200"]
    index => "logstash-%{+YYYY.MM.dd}"
  }
}
```

roles/nagios/defaults/main.yml

```
 GNU nano 7.2                                roles/nagios/defaults/main.yml
---
nagios_version: "4.4.14"
nagios_plugins_version: "2.4.6"

nagios_core_url: "https://assets.nagios.com/downloads/nagioscore/releases/nagios-{{ nagios_version }}.tar.gz"
nagios_plugins_url: "https://github.com/nagios-plugins/nagios-plugins/releases/download/release-{{ nagios_plugins_ve

nagios_install_dir: /usr/local/nagios
nagios_build_dir: /tmp

nagios_admin_user: charles-vence-espinas
nagios_admin_password: "Password123!"
```

roles/nagios/handlers/main.yml

```
 GNU nano 7.2                                roles/nagios/handlers/main.yml
---
- name: restart apache
  ansible.builtin.service:
    name: "{{ apache_service }}"
    state: restarted

- name: restart nagios
  ansible.builtin.service:
    name: nagios
    state: restarted
```

roles/nagios/tasks/build.yml

```
GNU nano 7.2                                    roles/nagios/tasks/build.yml
---
#Nagios Core
- name: Download Nagios Core tarball
  ansible.builtin.get_url:
    url: "{{ nagios_core_url }}"
    dest: "{{ nagios_build_dir }}/nagios-{{ nagios_version }}.tar.gz"
    mode: "0644"

- name: Extract Nagios Core source
  ansible.builtin.unarchive:
    src: "{{ nagios_build_dir }}/nagios-{{ nagios_version }}.tar.gz"
    dest: "{{ nagios_build_dir }}"
    remote_src: true
    creates: "{{ nagios_build_dir }}/nagios-{{ nagios_version }}/configure"

- name: Compile and install Nagios Core
  ansible.builtin.shell: |
    set -e
    ./configure --with-httpd-conf={{ apache_conf_dir }}
    make all
    make install-groups-users
    make install
    make install-daemoninit
    make install-commandmode
    make install-config
    make install-webconf
  args:
    chdir: "{{ nagios_build_dir }}/nagios-{{ nagios_version }}"
```

```
- name: Compile and install Nagios Core
  ansible.builtin.shell: |
    set -e
    ./configure --with-httpd-conf={{ apache_conf_dir }}
    make all
    make install-groups-users
    make install
    make install-daemoninit
    make install-commandmode
    make install-config
    make install-webconf
  args:
    chdir: "{{ nagios_build_dir }}/nagios-{{ nagios_version }}"
    creates: "{{ nagios_install_dir }}/bin/nagios"
  register: nagios_core_build
  notify:
    - restart apache
    - restart nagios
```

```
rgs:
  chdir: "{{ nagios_build_dir }}/nagios-plugins-{{ nagios_plugins_version }}"
  creates: "{{ nagios_install_dir }}/libexec/check_ping"
otify: restart nagios
```

roles/nagios/tasks/deps.yml

```
GNU nano 7.2                                    roles/nagios/tasks/deps.yml
-
ame: Update apt cache (Debian family)
nsible.builtin.apt:
  update_cache: true
  cache_valid_time: 3600
hen: ansible_os_family == "Debian"

ame: Install dnf-plugins-core (RedHat family)
nsible.builtin.dnf:
  name: dnf-plugins-core
  state: present
hen: ansible_os_family == "RedHat"

ame: Check whether the CRB repository is defined (RedHat family)
nsible.builtin.shell: dnf repolist --all | grep -q '^crb '
egister: crb_check
hanged_when: false
ailed_when: false
hen: ansible_os_family == "RedHat"

ame: Define the CRB repository if the VM does not have it (RedHat family)
nsible.builtin.yum_repository:
  name: crb
  description: CentOS Stream $releasever - CRB
  metalink: "https://mirrors.centos.org/metalink?repo=centos-crb-$stream&arch=$basearch&protocol=https,ht
  gpgcheck: true
  gpgkey: file:///etc/pki/rpm-gpg/RPM-GPG-KEY-centosofficial
  enabled: true
                                                    [ Read 42 lines ]
```

```
- name: Enable the CRB repository if it already exists (RedHat family)
  ansible.builtin.command: dnf config-manager --set-enabled crb
  changed_when: false
  when:
    - ansible_os_family == "RedHat"
    - crb_check.rc == 0

- name: Install build dependencies
  ansible.builtin.package:
    name: "{{ nagios_packages }}"
    state: present
```

roles/nagios/tasks/main.yml

GNU nano 7.2 roles/nagios/tasks/main.yml

```yaml
---
- name: Load OS-specific variables
  ansible.builtin.include_vars: "{{ ansible_os_family }}.yml"

- name: Install build dependencies
  ansible.builtin.include_tasks: deps.yml

- name: Open firewall and relax SELinux (RedHat family only)
  ansible.builtin.include_tasks: redhat.yml
  when: ansible_os_family == "RedHat"

- name: Build and install Nagios Core and Plugins
  ansible.builtin.include_tasks: build.yml

- name: Configure Apache, web login and services
  ansible.builtin.include_tasks: web.yml
```

roles/nagios/tasks/redhat.yml

GNU nano 7.2 roles/nagios/tasks/redhat.yml

```yaml
---
- name: Gather service facts
  ansible.builtin.service_facts:

- name: Allow HTTP through firewalld
  ansible.posix.firewalld:
    service: http
    permanent: true
    immediate: true
    state: enabled
  when:
    - ansible_facts.services['firewalld.service'] is defined
    - ansible_facts.services['firewalld.service'].state == 'running'

- name: Set SELinux to permissive (lab setting so Nagios CGI scripts work)
  ansible.posix.selinux:
    policy: targeted
    state: permissive
  when: ansible_selinux.status | default('disabled') == 'enabled'
```

roles/nagios/tasks/web.yml

```
  GNU nano 7.2                                  roles/nagios/tasks/web.yml
---
- name: Enable Apache modules rewrite and cgi (Debian family)
  ansible.builtin.command: "a2enmod {{ item }}"
  loop:
    - rewrite
    - cgi
  register: a2enmod_result
  changed_when: "'already enabled' not in a2enmod_result.stdout"
  notify: restart apache
  when: ansible_os_family == "Debian"

- name: Add the Apache user to the nagios group
  ansible.builtin.user:
    name: "{{ apache_user }}"
    groups: nagios
    append: true
  notify: restart apache

- name: Create the Nagios web admin login
  ansible.builtin.command: >
    htpasswd -bc {{ nagios_install_dir }}/etc/htpasswd.users
    {{ nagios_admin_user }} {{ nagios_admin_password | quote }}
  args:
    creates: "{{ nagios_install_dir }}/etc/htpasswd.users"
  no_log: true

- name: Enable and start Apache and Nagios
  ansible.builtin.service:
```

```
- name: Enable and start Apache and Nagios
  ansible.builtin.service:
    name: "{{ item }}"
    state: started
    enabled: true
  loop:
    - "{{ apache_service }}"
    - nagios
```

roles/nagios/vars/Debian.yml

```
  GNU nano 7.2                                  roles/nagios/vars/Debian.yml
---
apache_service: apache2
apache_user: www-data
apache_conf_dir: /etc/apache2/sites-enabled

nagios_packages:
  - autoconf
  - build-essential
  - gcc
  - make
  - libc6
  - wget
  - tar
  - unzip
  - bc
  - gawk
  - dc
  - gettext
  - apache2
  - apache2-utils
  - php
  - libapache2-mod-php
  - libgd-dev
  - openssl
  - libssl-dev
  - snmp
  - libnet-snmp-perl
```

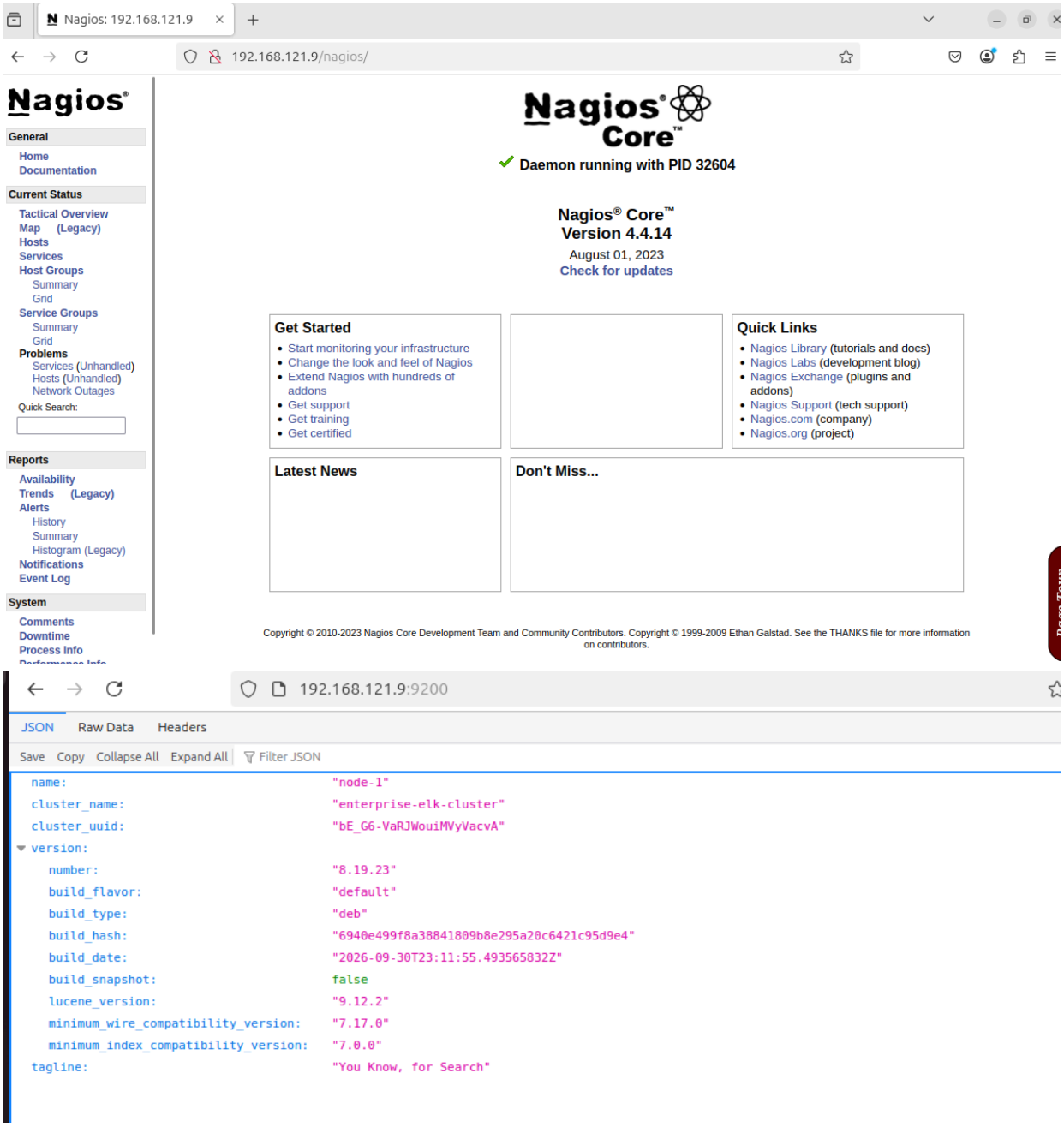
roles/nagios/vars/RedHat.yml

```
  GNU nano 7.2                                  roles/nagios/vars/RedHat.yml
---
apache_service: httpd
apache_user: apache
apache_conf_dir: /etc/httpd/conf.d

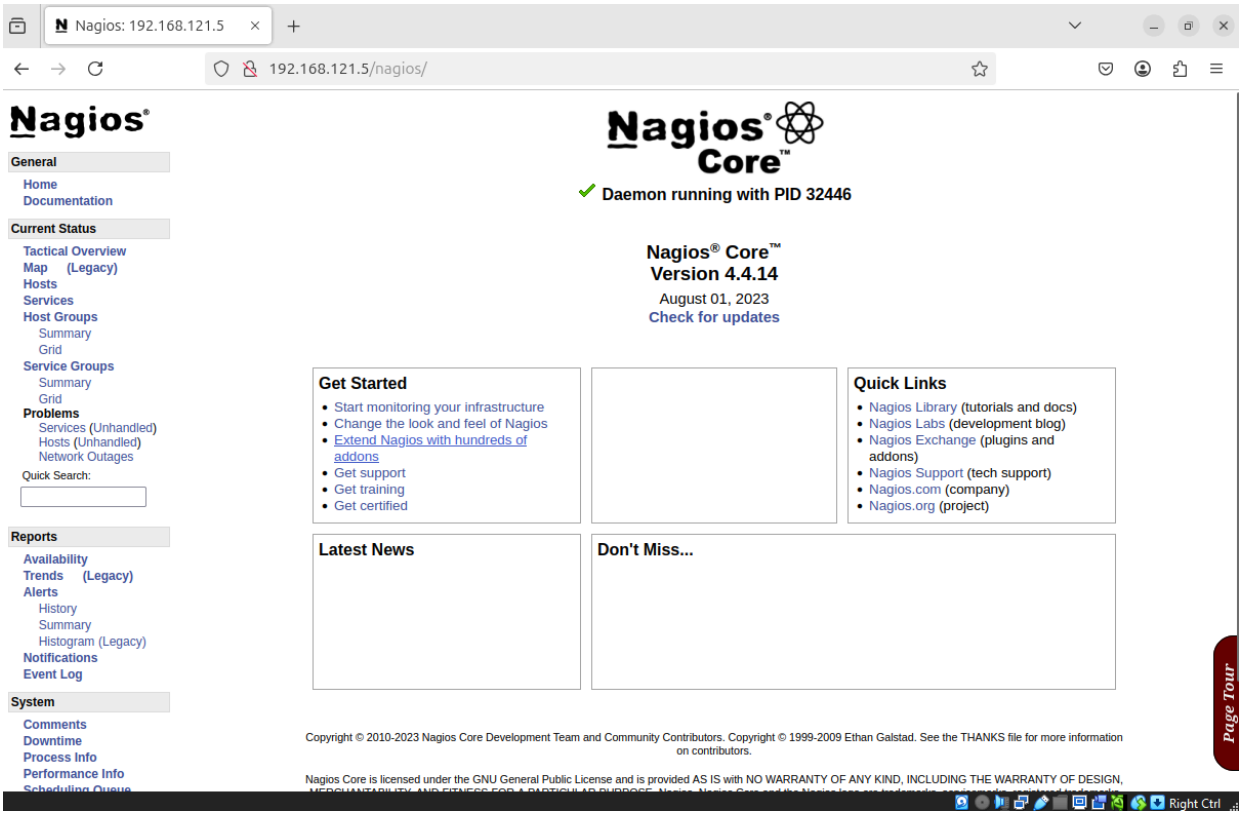
nagios_packages:
  - gcc
  - glibc
  - glibc-common
  - make
  - gettext
  - automake
  - autoconf
  - wget
  - tar
  - unzip
  - bc
  - gawk
  - openssl-devel
  - net-snmp
  - net-snmp-utils
  - httpd
  - httpd-tools
  - php
  - gd
  - gd-devel
  - perl
  - python3-libselinux
                                                  [ Read 28 lines ]
```

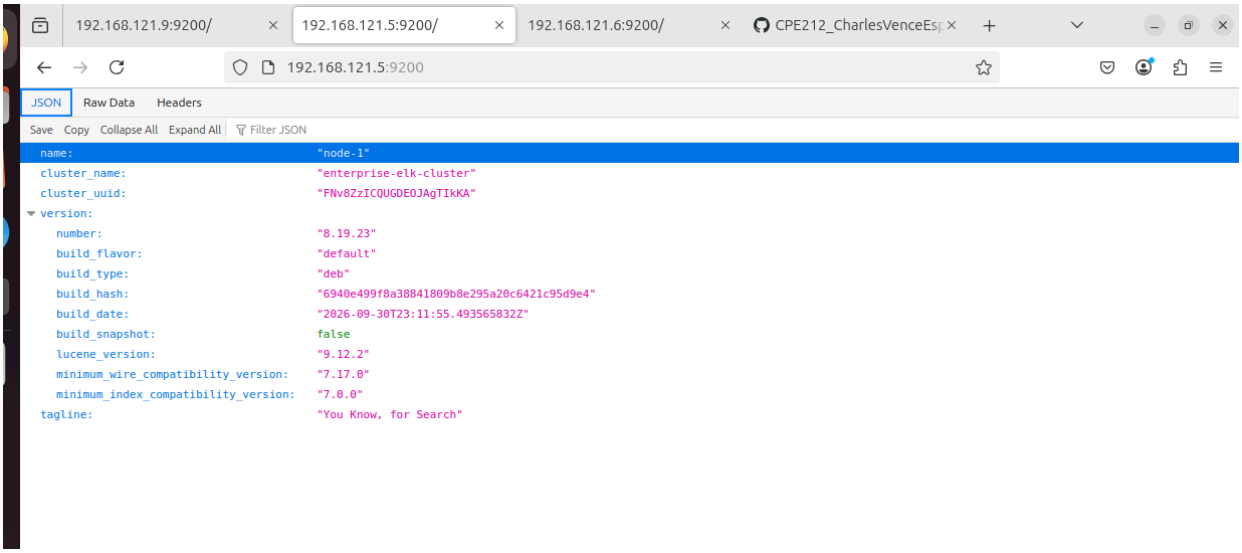
Verification

server1

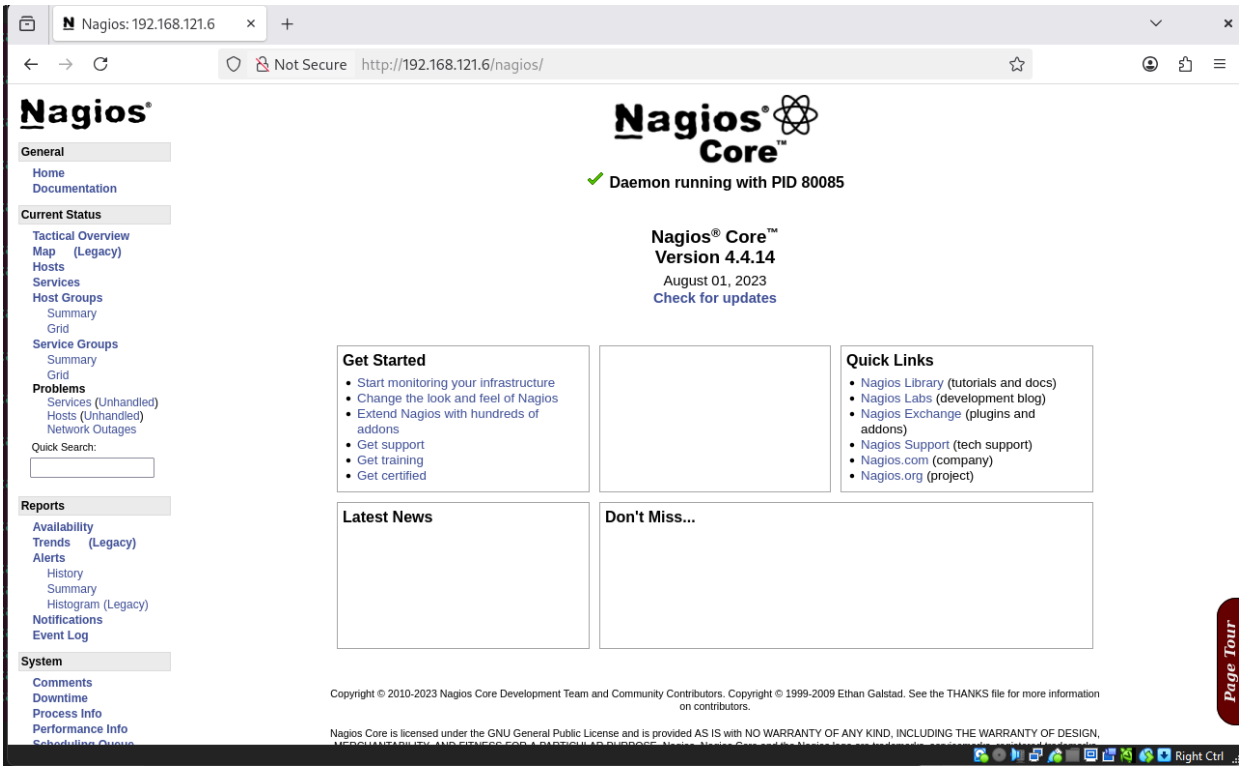

Nagios: 192.168.121.9
192.168.121.9/nagios/
Nagios
General
Home
Documentation
Current Status
Tactical Overview
Map (Legacy)
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services (Unhandled)
Hosts (Unhandled)
Network Outages
Quick Search:
Reports
Availability
Trends (Legacy)
Alerts
History
Summary
Histogram (Legacy)
Notifications
Event Log
System
Comments
Downtime
Process Info
Nagios Core
Daemon running with PID 32604
Nagios® Core™
Version 4.4.14
August 01, 2023
Check for updates
Get Started
Start monitoring your infrastructure
Change the look and feel of Nagios
Extend Nagios with hundreds of addons
Get support
Get training
Get certified
Quick Links
Nagios Library (tutorials and docs)
Nagios Labs (development blog)
Nagios Exchange (plugins and addons)
Nagios Support (tech support)
Nagios.com (company)
Nagios.org (project)
Latest News
Don't Miss...
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192.168.121.9:9200
JSON
Raw Data
Headers
Save
Copy
Collapse All
Expand All
Filter JSON
name: "node-1"
cluster_name: "enterprise-elk-cluster"
cluster_uuid: "bE_G6-VaRJWouiMVyVacvA"
version:
number: "8.19.23"
build_flavor: "default"
build_type: "deb"
build_hash: "6940e499f8a38841809b8e295a20c6421c95d9e4"
build_date: "2026-09-30T23:11:55.493565832Z"
build_snapshot: false
lucene_version: "9.12.2"
minimum_wire_compatibility_version: "7.17.0"
minimum_index_compatibility_version: "7.0.0"
tagline: "You Know, for Search"

serve2


Nagios: 192.168.121.5
192.168.121.5/nagios/
Nagios
General
Home
Documentation
Current Status
Tactical Overview
Map (Legacy)
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services (Unhandled)
Hosts (Unhandled)
Network Outages
Quick Search:
Reports
Availability
Trends (Legacy)
Alerts
History
Summary
Histogram (Legacy)
Notifications
Event Log
System
Comments
Downtime
Process Info
Performance Info
Nagios Core
Daemon running with PID 32446
Nagios® Core™
Version 4.4.14
August 01, 2023
Check for updates
Get Started
Start monitoring your infrastructure
Change the look and feel of Nagios
Extend Nagios with hundreds of addons
Get support
Get training
Get certified
Quick Links
Nagios Library (tutorials and docs)
Nagios Labs (development blog)
Nagios Exchange (plugins and addons)
Nagios Support (tech support)
Nagios.com (company)
Nagios.org (project)
Latest News
Don't Miss...
Page Tour
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Right Ctrl


192.168.121.9:9200/
192.168.121.5:9200/
192.168.121.6:9200/
CPE212_CharlesVenceEs
192.168.121.5:9200
JSON
Raw Data
Headers
Save
Copy
Collapse All
Expand All
Filter JSON
name: "node-1"
cluster_name: "enterprise-elk-cluster"
cluster_uuid: "FNv8ZzICQUGDEOJAgTIkKA"
version:
number: "8.19.23"
build_flavor: "default"
build_type: "deb"
build_hash: "6940e499f8a38841809b8e295a20c6421c95d9e4"
build_date: "2026-09-30T23:11:55.493565832Z"
build_snapshot: false
lucene_version: "9.12.2"
minimum_wire_compatibility_version: "7.17.0"
minimum_index_compatibility_version: "7.0.0"
tagline: "You Know, for Search"

server3


Nagios: 192.168.121.6
Not Secure
http://192.168.121.6/nagios/
Nagios
General
Home
Documentation
Current Status
Tactical Overview
Map (Legacy)
Hosts
Services
Host Groups
Summary
Grid
Service Groups
Summary
Grid
Problems
Services (Unhandled)
Hosts (Unhandled)
Network Outages
Quick Search:
Reports
Availability
Trends (Legacy)
Alerts
History
Summary
Histogram (Legacy)
Notifications
Event Log
System
Comments
Downtime
Process Info
Performance Info
Nagios Core
Daemon running with PID 80085
Nagios® Core™
Version 4.4.14
August 01, 2023
Check for updates
Get Started
Start monitoring your infrastructure
Change the look and feel of Nagios
Extend Nagios with hundreds of addons
Get support
Get training
Get certified
Quick Links
Nagios Library (tutorials and docs)
Nagios Labs (development blog)
Nagios Exchange (plugins and addons)
Nagios Support (tech support)
Nagios.com (company)
Nagios.org (project)
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Right Ctrl


192.168.121.9:9200/
192.168.121.5:9200/
192.168.121.6:9200/
CPE212_CharlesVenceEs
192.168.121.6:9200
JSON
Raw Data
Headers
Save
Copy
Collapse All
Expand All
Filter JSON
name: "node-1"
cluster_name: "enterprise-elk-cluster"
cluster_uuid: "y1zzSZRbQ4a4UwwJZs-a0w"
version:
number: "8.19.23"
build_flavor: "default"
build_type: "rpm"
build_hash: "6940e499f8a38841809b8e295a20c6421c95d9e4"
build_date: "2026-09-30T23:11:55.493565832Z"
build_snapshot: false
lucene_version: "9.12.2"
minimum_wire_compatibility_version: "7.17.0"
minimum_index_compatibility_version: "7.0.0"
tagline: "You Know, for Search"

Execution

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/Final_Exam_Espinas$ ansible-playbook -i inven
tory.yaml site.yml -K
BECOME password:

PLAY [Install and Configure Nagios Core across Ubuntu and CentOS] **************

TASK [Gathering Facts] *********************************************************
ok: [server1]
ok: [server2]
ok: [centOS]

TASK [nagios : Load OS-specific variables] *************************************
ok: [server1]
ok: [server2]
ok: [centOS]

TASK [nagios : Install build dependencies] *************************************
included: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/Final_Exam_Espinas/roles/nagios/tasks/deps.
yml for server1, server2, centOS

TASK [nagios : Update apt cache (Debian family)] *******************************
skipping: [centOS]
ok: [server1]
ok: [server2]

TASK [nagios : Install dnf-plugins-core (RedHat family)] ***********************
skipping: [server1]
skipping: [server2]
ok: [centOS]

TASK [nagios : Check whether the CRB repository is defined (RedHat family)] ****
skipping: [server1]
skipping: [server2]
ok: [centOS]

TASK [nagios : Define the CRB repository if the VM does not have it (RedHat family)] ***
skipping: [server1]
skipping: [server2]
skipping: [centOS]

TASK [nagios : Enable the CRB repository if it already exists (RedHat family)] ***
skipping: [server1]
skipping: [server2]
ok: [centOS]

TASK [nagios : Install build dependencies] *************************************
ok: [server1]
changed: [server2]
changed: [centOS]

TASK [nagios : Open firewall and relax SELinux (RedHat family only)] ***********
skipping: [server1]
skipping: [server2]
included: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/Final_Exam_Espinas/roles/nagios/tasks/redha
t.yml for centOS

TASK [nagios : Gather service facts] *******************************************
ok: [centOS]
```

```
TASK [nagios : Allow HTTP through firewalld] ***********************************
changed: [centOS]

TASK [nagios : Set SELinux to permissive (lab setting so Nagios CGI scripts work)] ***
changed: [centOS]

TASK [nagios : Build and install Nagios Core and Plugins] **********************
included: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/Final_Exam_Espinas/roles/nagios/tasks/build
.yml for server1, server2, centOS

TASK [nagios : Download Nagios Core tarball] ***********************************
changed: [server1]
changed: [centOS]
changed: [server2]

TASK [nagios : Extract Nagios Core source] *************************************
changed: [server1]
changed: [server2]
changed: [centOS]

TASK [nagios : Compile and install Nagios Core] ********************************
changed: [server2]
changed: [server1]
changed: [centOS]

TASK [nagios : Reload systemd so the nagios unit is picked up] *****************
ok: [server1]
ok: [server2]
ok: [centOS]
```

```
TASK [nagios : Download Nagios Plugins tarball] ********************************
changed: [server1]
changed: [server2]
changed: [centOS]

TASK [nagios : Extract Nagios Plugins source] **********************************
changed: [server1]
changed: [server2]
changed: [centOS]

TASK [nagios : Compile and install Nagios Plugins] *****************************
changed: [server1]
changed: [server2]
changed: [centOS]

TASK [nagios : Configure Apache, web login and services] ***********************
included: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/Final_Exam_Espinas/roles/nagios/tasks/web.y
ml for server1, server2, centOS

TASK [nagios : Enable Apache modules rewrite and cgi (Debian family)] **********
skipping: [centOS] => (item=rewrite)
skipping: [centOS] => (item=cgi)
skipping: [centOS]
changed: [server1] => (item=rewrite)
changed: [server2] => (item=rewrite)
changed: [server1] => (item=cgi)
changed: [server2] => (item=cgi)

TASK [nagios : Add the Apache user to the nagios group] ************************
changed: [server2]
changed: [server1]
changed: [centOS]
```

```
TASK [nagios : Create the Nagios web admin login] ******************************
changed: [server1]
changed: [centOS]
changed: [server2]

TASK [nagios : Enable and start Apache and Nagios] *****************************
ok: [server2] => (item=apache2)
ok: [server1] => (item=apache2)
changed: [server2] => (item=nagios)
changed: [server1] => (item=nagios)
changed: [centOS] => (item=httpd)
changed: [centOS] => (item=nagios)

RUNNING HANDLER [nagios : restart apache] **************************************
changed: [server1]
changed: [server2]
changed: [centOS]

RUNNING HANDLER [nagios : restart nagios] **************************************
changed: [server1]
changed: [centOS]
changed: [server2]

PLAY [Apply common configuration across all ELK nodes] *************************

TASK [Gathering Facts] *********************************************************
ok: [server1]
ok: [server2]
ok: [centOS]
```

```
TASK [common : Create 2GB swap file] ****************************************
changed: [server2]
changed: [server1]
changed: [centOS]

TASK [common : Set correct permissions on swap file] ***************************
changed: [server1]
changed: [server2]
changed: [centOS]

TASK [common : Make swap filesystem] ****************************************
ok: [server1]
ok: [centOS]
ok: [server2]

TASK [common : Enable swap] *************************************************
changed: [server2]
changed: [server1]
changed: [centOS]

TASK [common : Ensure swap is mounted on boot in fstab] ************************
changed: [server1]
changed: [server2]
changed: [centOS]

TASK [common : Install prerequisites on Debian/Ubuntu] *************************
skipping: [centOS]
changed: [server2]
changed: [server1]
```

```
TASK [common : Add Elastic GPG key on Debian/Ubuntu] ***************************
skipping: [centOS]
changed: [server1]
changed: [server2]

TASK [common : Add Elastic repository on Debian/Ubuntu] ************************
skipping: [centOS]
changed: [server2]
changed: [server1]

TASK [common : Import Elastic GPG key on RHEL/CentOS] **************************
skipping: [server1]
skipping: [server2]
changed: [centOS]

TASK [common : Add Elastic repository on RHEL/CentOS] **************************
skipping: [server1]
skipping: [server2]
changed: [centOS]

TASK [elasticsearch : Install Elasticsearch] ***********************************
changed: [server2]
changed: [server1]
changed: [centOS]
```

```
TASK [elasticsearch : Ensure Elasticsearch data and log directories exist] *****
changed: [server1] => (item=/var/lib/elasticsearch)
changed: [server2] => (item=/var/lib/elasticsearch)
changed: [server1] => (item=/var/log/elasticsearch)
changed: [server2] => (item=/var/log/elasticsearch)
changed: [centOS] => (item=/var/lib/elasticsearch)
changed: [centOS] => (item=/var/log/elasticsearch)

TASK [elasticsearch : Limit JVM heap size to 512MB] ****************************
changed: [server1]
changed: [centOS]
changed: [server2]

TASK [elasticsearch : Configure elasticsearch.yml] *****************************
changed: [server1]
changed: [centOS]
changed: [server2]

TASK [elasticsearch : Enable and start Elasticsearch service] ******************
changed: [server1]
changed: [server2]
changed: [centOS]

TASK [elasticsearch : Allow Elasticsearch port 9200 on Debian/Ubuntu (UFW)] ****
skipping: [centOS]
changed: [server1]
changed: [server2]

TASK [elasticsearch : Allow Elasticsearch port 9200 on RHEL/CentOS (firewalld)] ***
skipping: [server1]
skipping: [server2]
changed: [centOS]
```

```
RUNNING HANDLER [elasticsearch : restart elasticsearch] ***********************
changed: [server1]
changed: [server2]
changed: [centOS]

PLAY RECAP *********************************************************************
centOS                     : ok=40   changed=27   unreachable=0    failed=0    skipped=7    rescued=0    ignored=0
server1                    : ok=36   changed=26   unreachable=0    failed=0    skipped=8    rescued=0    ignored=0
server2                    : ok=36   changed=27   unreachable=0    failed=0    skipped=8    rescued=0    ignored=0
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/Final_Exam_Espinas$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

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Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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